

Place-field repetition is what a predictive map must do

when self-motion cannot resolve the symmetry

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The puzzle

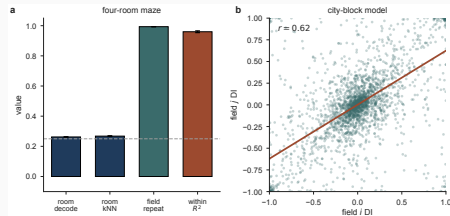
Place cells in identical compartments fire at the same relative location in every one.

“a high degree of spatial repetition”

Spiers et al., 2015

Usual reading: the hippocampus fails to tell the rooms apart.

We show the opposite.



A network trained only to predict its next observation does not represent physical space X .

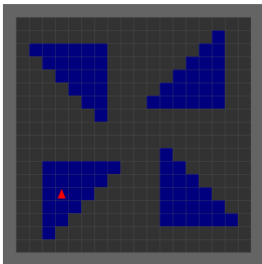
It represents the quotient X/G .

- G is the subgroup that self-motion fails to break.
- Set by equivariance, not by information content.

- **Orbit:** a set of positions related by the symmetry, e.g. $\{x, R^2x\}$ for a 180° rotation.
- **Phase:** which member of the orbit produced the current state.
- **Quotient** X/G : physical space X with every orbit collapsed to one point.
- **Invariant:** unchanged under the group, $\phi(h + g) = \phi(h)$.
- **Orbit-phase accuracy:** can a decoder tell which orbit member produced a given hidden state; chance is $1/|G|$.

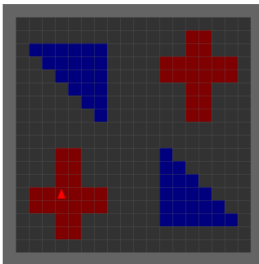
project5_symmetry: new arena layouts

S4: C4 rotational symmetry



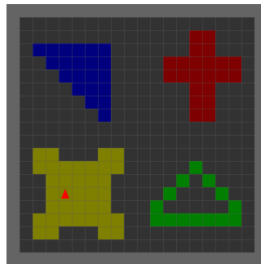
H2 mean=5.54 | landmark tiles=84

S2: 180-degree symmetry



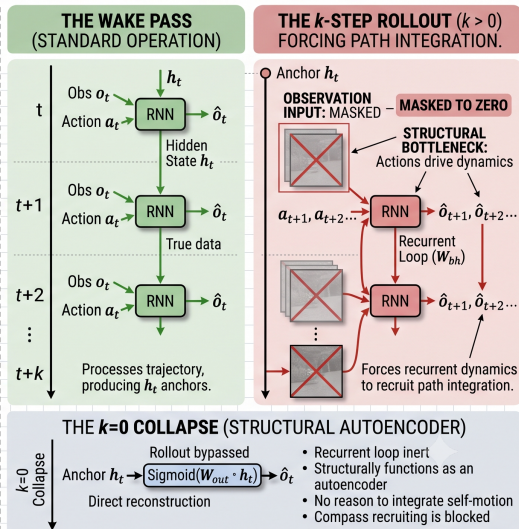
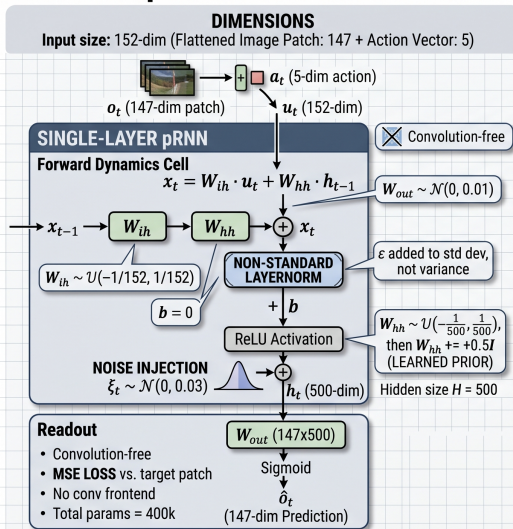
H2 mean=3.21 | landmark tiles=86

S1: asymmetric quadrants

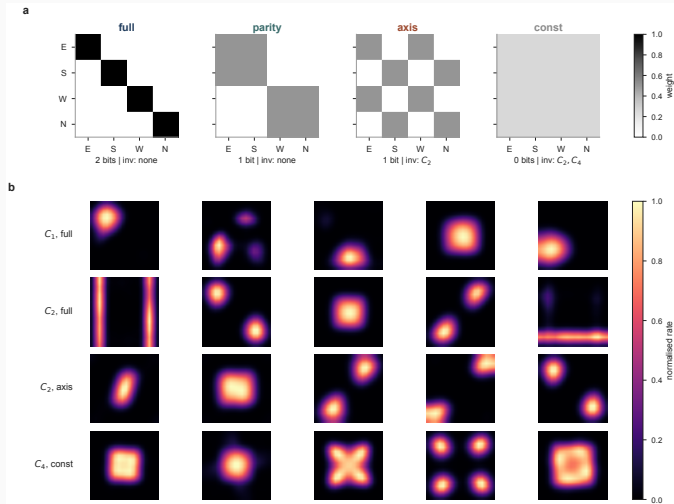


H2 mean=2.15 | landmark tiles=94

pRNN CORE ARCHITECTURE & ROLLOUT DYNAMICS



The manipulation

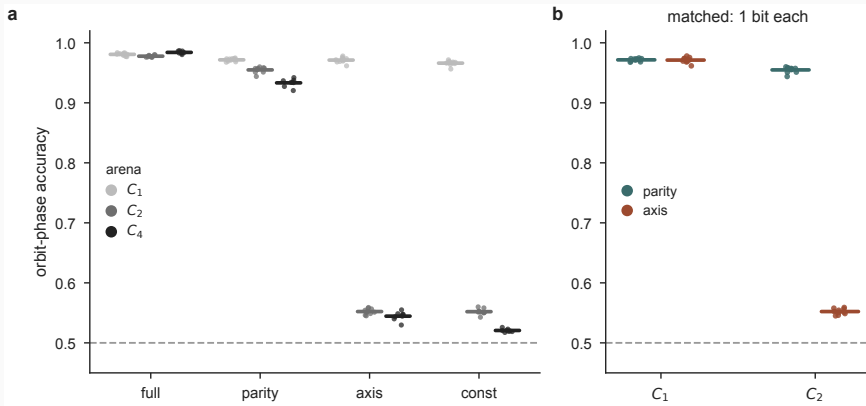


Folding onto the quotient

If the environment has a symmetry G and the head-direction encoding is G -invariant, the posterior over position given any history of observations is G -invariant, and no decoder of the orbit element can beat $1/|G|$.

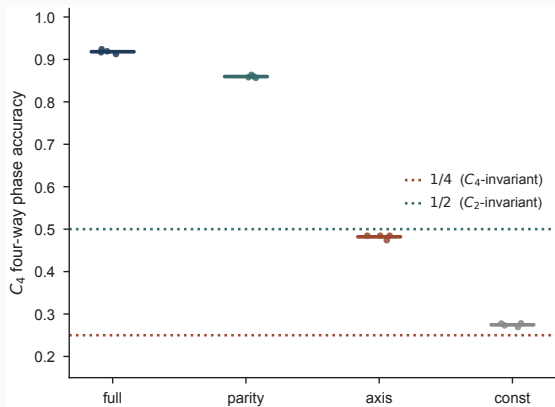
An upper bound, not a prediction. Whether a trained network reaches it is empirical.

Invariance, not information, folds the code



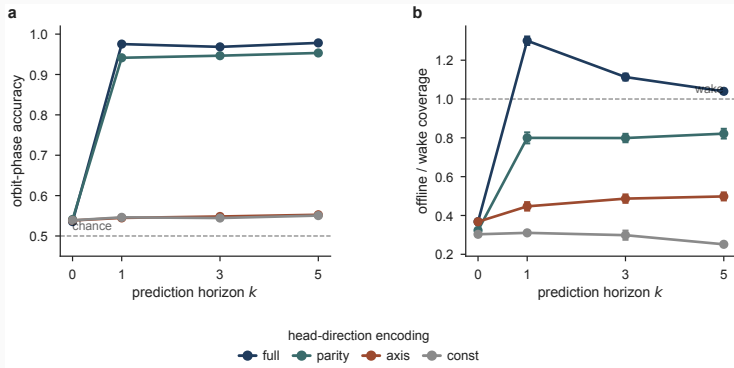
(a) Each dot is one trained network; can a decoder tell which of two 180° -related positions produced this hidden state? *full/parity* stay near 1; *axis/const* collapse to chance, only in the symmetric arenas. (b) *axis* and *parity* carry exactly the same one bit. Indistinguishable in C_1 ; completely separated in C_2 . The only difference between them is which one is invariant to the rotation.

The C_4 ceiling is exactly $1/|G|$



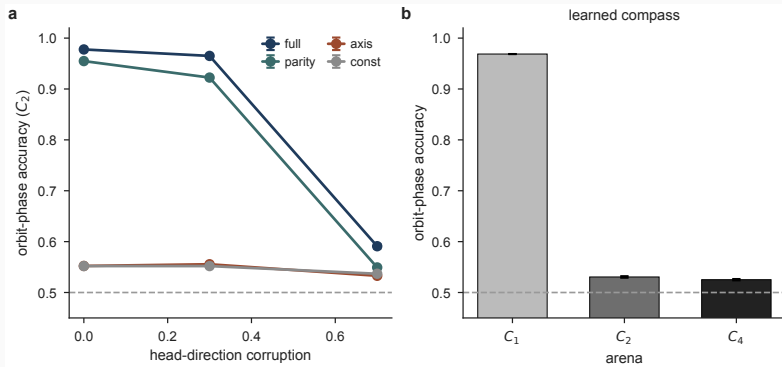
Same test, now decoding among 4 rotational copies (chance $1/4$). *axis* is invariant only under 180° , so it lands on the $1/2$ ceiling; *const* is invariant under the full 90° rotation, so it lands on $1/4$. The dotted ceilings are computed from group theory before seeing the data, not fit to it.

The fold requires prediction



x-axis is the prediction horizon k : how many steps ahead the network is trained to predict. (a) At $k = 0$ (predict the current frame, a pure autoencoder) every encoding sits at chance. At $k \geq 1$ they jump apart and never move again: a step, not a gradient. (b) The same step in offline replay coverage.

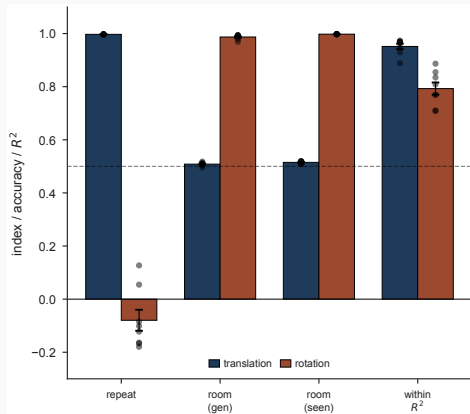
Robust, and unaided by any hand-built encoding



(a) x-axis is what fraction of heading steps are randomly corrupted. *full*/*parity* degrade smoothly as the compass gets noisier; *axis*/*const* stay pinned at chance throughout, since corruption doesn't change whether the signal is invariant. (b) A network given only angular velocity, no absolute heading, still resolves the asymmetric arena and still folds the symmetric one, with no invariant encoding built in by hand.

The same law explains the animal data

Neural



Translated compartments: fields repeat, room undecodable.
Rotated compartments: repetition vanishes, room decodes perfectly. Grieves et al., 2016.

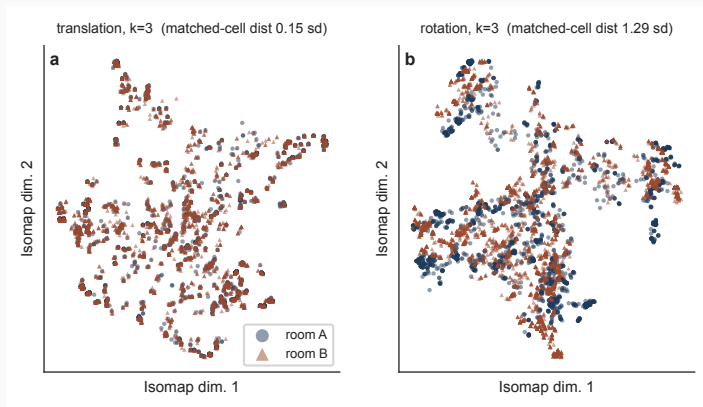
Behavioral

A separate cohort learned to tell the rooms apart.

Learning was significantly impaired exactly where the neural code folds.

“place field repetition constrains spatial learning”

The fold, seen geometrically



Isomap embedding, no decoder involved (Levenstein et al. style). Colour is room identity, never shown to Isomap. Matched-cell distance in the embedding: 0.13 vs. 1.31 (embedding-spread units), $n = 8$ each, $p = 1.6 \times 10^{-4}$.

- The belief process descends to a Markov process on X/G : a MDP homomorphism.
- States that share the same future are bisimilar; this is a bisimulation relation.
- A predictive learner acquiring the successor representation of its belief state should represent that quotient, with no reward and no group given to the network.

- A predictive map represents a quotient X/G , not physical space X .
- Proved as a ceiling, saturated only under a predictive objective.
- One principle explains repetition, neurally and behaviorally.
- One architecture, one real-neuron test that is consistent with the quotient, not proof of it.

A cognitive map represents space only up to the symmetries its own motion cannot break.

You already built the compass this theory needs

LaChance & Hasselmo, *Nat Commun* **15**:8025 (2024). Two identical cue cards, on opposite walls of a square — a symmetry the visual scene cannot break.

head-direction cells	become bidirectional?	
postrhinal (POR)	yes	$p = 2 \times 10^{-10}$
retrosplenial (RSC)	yes	$p = 1.4 \times 10^{-4}$
entorhinal (MEC/PaS)	no	$p = 0.62$

A bidirectional head-direction cell *is* a C_2 -invariant compass: it reports the same heading at x and at $R_{180}x$.

Zhang, Grieves & Jeffery (2022) get the same object by varying the environment: RSC tuning is unidirectional in a circle, **bidirectional** in a C_2 box, **tetradirectional** in a C_4 box.

Neither paper recorded a single place cell.

We have that compass, and it is the *only* thing that matters

Orbit-phase decoding: can a decoder say *which* of two 180° -related positions the animal occupies? Chance = 0.500. One dot per network, $n \approx 10$.

arena	<i>full</i>	<i>parity</i>	<i>axis</i>	<i>const</i>
C_1 (no symmetry)	0.981	0.972	0.971	0.966
C_2	0.978	0.955	0.552	0.552
C_4	0.984	0.933	0.544	0.521

axis = $\{E, W\}$ vs $\{N, S\}$ — **invariant** under 180° .

parity = $\{E, S\}$ vs $\{W, N\}$ — **breaks** 180° .

Both carry exactly one bit of heading. Same information, same architecture, same walls, same corners, same boundary cells. Opposite folds.

No boundary-vector model can state this: it has no representation of *which* bit the compass carries. Nor can any animal experiment run it.

It is the objective, not the assumptions

Same network, same compass, same inputs. Only the rollout horizon k changes. At $k = 0$ the target is the network's own current observation — a pure autoencoder.

k	<i>full</i>	<i>parity</i>	<i>axis</i>	<i>const</i>	<i>axis vs parity</i>
0 (autoencoder)	0.536	0.541	0.539	0.539	$p = 0.59$
1	0.975	0.941	0.545	0.546	$p = 0.002$
5	0.978	0.953	0.553	0.551	$p = 0.002$

At $k = 0$ the compass is *right there* — the input alone separates 1228 of 1296 states — and the network **discards it**. All four encodings sit at chance. *axis* and *parity* are indistinguishable.

One step of prediction makes heading load-bearing.

Reconstructing what you see never requires knowing which of two identical rooms you are in. Predicting what you will see next does.

The experiment, and it is yours to run

Record place cells in the square, with **two identical opposing cue cards**.

Prediction, no free parameters:

- C_2 place-field repetition — fields recur at 180° -related locations.
- A decoder for *which half* of the arena the animal occupies is bounded at $\frac{1}{2}$.

Why it is a real test: your MEC head-direction cells did *not* go bidirectional ($p = 0.62$) and your grid cells did not change. If the hippocampus reads only that channel, nothing happens — and we are wrong. But POR's densest hippocampal projection is to *subiculum*, and the LMN lesion (Harland 2017) *does* fold hippocampal fields.

The apparatus already exists. A null falsifies us.